

INTEGRATED PROJECT REPORT

On

SMART ANTI-PROXY ATTENDANCE SYSTEM

Submitted in partial fulfilment of the requirement for the Course
Integrated Project (22CS038)

Of

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CERTIFICATE

This is to be certified that the project entitled "**Smart Anti-Proxy Attendance System**" has been submitted for the Bachelor of Computer Science Engineering at Chitkara University, Punjab during the academic semester January 2026 - May 2026 is a bonafide piece of project work carried out by **Mahak Choudhary (2310991973), Mehak (2310991983), and Mishti Kharra (2310991987)** towards the partial fulfilment for the award of the course Integrated Project (22CS038) under the guidance of **Dr. Rajdavinder Singh Boparai** and supervision.

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CANDIDATE'S DECLARATION

We, **Mahak Choudhary (2310991973), Mehak (2310991983), and Mishti Kharra (2310991987)**, B.E.-2023 of the Chitkara University, Punjab hereby declare that the Integrated Project (22CS038) Project Report entitled "**SMART ANTI-PROXY ATTENDANCE SYSTEM**" is an original work and data provided in the study is authentic to the best of our knowledge. This report has not been submitted to any other Institute for the award of any other course.

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ABSTRACT

The Smart Anti-Proxy Attendance System is an Android-based application designed to prevent fake or proxy attendance in academic institutions. Traditional attendance methods such as paper registers or static QR codes cannot verify whether a student is physically present throughout the class. This system solves that problem using a secure three-layer verification framework: device authentication, real-time geofencing, and continuous location monitoring.

After login through Firebase, the application retrieves the device's unique `ANDROID_ID` using the `Android Settings`. Secure framework. This ID is linked with the student's profile in Cloud Firestore, ensuring that one account can only be used on one registered device and preventing credential sharing.

For attendance verification, the system uses the Google Play Services `FusedLocationProviderClient` API to access live GPS coordinates. When the teacher generates a random 4-digit attendance code, the application checks the student's distance from the instructor's location using `Location.distanceBetween()`. Attendance is marked only if the student is within a secure 50-meter geofence radius.

To ensure students remain in class after marking attendance, the system runs a persistent Android Foreground Service (`AttendanceTrackingService`). This background service verifies the student's location every two minutes. If the student remains outside the allowed geofence for 5 consecutive checks (10 minutes), the system automatically removes the attendance record from the database.

The proposed system provides a scalable, cost-effective, and reliable attendance solution without requiring biometric hardware. It improves attendance authenticity, reduces manual work, and ensures continuous student presence monitoring.

Keywords: Anti-Proxy Attendance, `ANDROID_ID`, Geofencing, Foreground Service, GPS Tracking, Firebase Authentication, Cloud Firestore, Continuous Monitoring.

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CHAPTER 1

INTRODUCTION TO THE PROJECT

Modern educational institutions place significant emphasis on attendance monitoring as a core metric for academic discipline, student engagement, and regulatory compliance. However, traditional methodologies utilized for capturing attendance continue to struggle with inefficiencies and vulnerability to manipulation. As universities scale, managing lecture logbooks and manual tracking becomes a resource-heavy operation that takes away valuable time from actual instruction. To address these systemic inefficiencies, this project introduces an automated, smartphone-driven architectural solution designed to ensure absolute accountability in attendance logging.

1.1. Background

For decades, institutional attendance tracking has relied on traditional manual roll calls, physical sign-in registers, or basic semi-automated digital structures such as broadcasted numerical PINs and static QR codes[6]. While these conventional methodologies are straightforward to deploy, they exhibit critical structural vulnerabilities that compromise database integrity. The most prevalent issue is "proxy attendance," a malpractice where present students validate identity transactions on behalf of absent peers[6].

With the emergence of mobile-first academic systems, institutions have attempted to patch these flaws by integrating web portals and standalone client apps. However, standard software configurations are easily bypassed; for instance, a student can easily transmit a screen capture of a classroom QR code or message a session PIN via instant messaging channels to off-site peers, allowing remote check-ins. Furthermore, existing architectures fail to monitor presence after the initial transaction checkpoint. This design flaw permits individuals to leave the classroom perimeter immediately after validation, creating deceptive records that undermine administrative discipline and compliance metrics.

1.2. Problem Statement

The primary challenge in modern institutional tracking is maintaining the authenticity and real-time accuracy of attendance records. Traditional frameworks waste instructional time and lack defenses against logical identity manipulation. The system addresses three primary technical vulnerabilities:

- **The Multi-Account Proxy Loophole:** Absent individuals regularly share active authentication credentials or rolling session keys with peers inside the room. This allows a single device to execute multiple sequential check-in writes, masking absenteeism[6].
- **Post-Validation Perimeter Desertion:** Conventional location-bounded platforms validate coordinates only at the entry checkpoint. This enables a loophole where students pass the initial geofence, commit the database write, and instantly exit the room while receiving full attendance credit[6].
- **Credential Proliferation via Visual Media:** Static optical or text tokens are highly vulnerable to unauthorized sharing over digital channels, enabling remote, fraudulent presence logging[6].

To resolve these security exploits entirely, the developed system enforces a multi-tier client-server Android architecture that executes three mandatory verification layers:

- **Device Hardware Binding:** To neutralize credential sharing, the client application queries the Android operating system at runtime to extract the node's unique hardware signature using the Settings.Secure.ANDROID_ID token[5]. This key is permanently bound to the user profile schema within Cloud Firestore during the primary login instance[7]. Future authentication requests execute strict runtime checks against this baseline; mismatches terminate the login transaction, successfully enforcing a strict 1:1 account-to-device infrastructure limit[6].
- **Real-Time Proximity Geo-Fencing:** To ensure absolute presence, the instructor terminal aggregates current coordinates via the FusedLocationProviderClient framework when publishing a lecture[1]. Upon token submission, the student application queries live device location coordinates and invokes the native framework method Location.distanceBetween() to compute linear displacement vectors[4]. Transactions exceeding the rigorous **50-meter geo-fence radius** are rejected by database security parameters[2].

- **Continuous Foreground Monitoring:** To eliminate early departures, a successful check-in initializes a persistent background daemon via the persistent **AttendanceTrackingService**[3]. This component evaluates device location vectors precisely every 2 minutes. Every failed boundary check increments a tracking failure counter inside the system database[7]. If a user logs **5 consecutive geofence failures** (signaling a continuous 10-minute classroom exit), the background service bypasses field modifications and executes an explicit `.delete()` database operation, completely purging the attendance transaction log for that active session[2].

CHAPTER 2

SOFTWARE AND HARDWARE REQUIREMENT SPECIFICATION

To ensure the seamless development, deployment, and execution of the Smart Anti-Proxy Attendance System, a robust specification of technical requirements and underlying methodologies is essential. This chapter outlines the structural engineering methods applied to tackle the proxy problem, the comprehensive development environment configured on the workstations, and the fundamental hardware and software execution baselines required by both clients and administrative nodes to run the application effectively.

2.1. Methods

The core engineering approach of this system relies on a synchronized three-layer validation protocol to verify student presence. The systematic flow is divided into three interconnected technical methods:

- **Hardware-Level Device Binding:** Upon initial authentication via Firebase Authentication, the client application invokes the Android Settings.Secure framework to extract the smartphone's unique alphanumeric identifier (ANDROID_ID)[5]. This token is mapped permanently to the student's profile document schema within Cloud Firestore[7]. Every subsequent authentication instance triggers an interceptor routine that compares the physical node's current hardware signature against the registered cloud baseline; any mismatch immediately blocks database access, restricting one student account to one physical smartphone[6].
- **Dynamic Geo-Fencing Proximity Validation:** The system utilizes the Google Play Services FusedLocationProviderClient API to aggregate high-accuracy, real-time geographical coordinates[1]. When an instructor initiates a classroom session via TeacherActivity.java by publishing a randomized 4-digit code, the student application calculates the exact linear distance between the student's live coordinates and the instructor's baseline coordinates using the native framework method Location.distanceBetween(). The transaction is committed to the Firestore database only if the calculated spatial distance satisfies the strict 50-meter geo-fence radius.

- **Continuous Foreground Presence Monitoring:** To prevent post-validation classroom departures after the initial check-in, the application immediately provisions an un-killable Android Foreground Service (AttendanceTrackingService) with a persistent system notification[3]. This service executes a continuous, resource-optimized background Handler polling loop every 120,000 milliseconds (2 minutes) to verify geofence compliance[3]. If a user logs 5 consecutive geofence failures (a continuous 10-minute breach), the background engine triggers an automated database purge by executing an explicit .delete() command on the active attendance document, stripping the user's presence log[2].

2.2. Programming/Working Environment

The development, compilation, and testing pipeline was constructed leveraging professional open-source integrated development environments (IDEs) and cloud architectures, maintaining an optimized institutional resource footprint. The core development infrastructure consists of:

- **Primary Integrated Development Environment (IDE):** Android Studio (Hedgehog / Iguana or latest stable release) integrated with the Android SDK, utilizing Java (JDK 17) under strict Object-Oriented Programming (OOP) principles for activity lifecycle management, database callback parsing, and background service orchestration[3].
- **Backend Cloud Architecture:** Google Firebase Ecosystem. Firebase Authentication orchestrates secure, encrypted email-based user access tokens. Cloud Firestore serves as the live, real-time NoSQL document-oriented database layer, managing concurrent check-in transactions, matching ANDROID_ID hardware keys, and updating ongoing classroom tracking sessions[7].
- **Workstation Operating System Environment:** Microsoft Windows 11 / macOS environments utilized for drafting programmatic logic, building application packages (APKs), and profiling memory/location execution threads.

2.3. Requirements to Run the Application

The application architecture divides execution baselines into precise workstation and runtime mobile environment standards:

2.3.1 Hardware Requirements:

- **Developer/Host Workstation:** A host workstation equipped with an Intel i5 / AMD Ryzen 5 or Apple M-series processor, a minimum of 8GB physical RAM (16GB recommended for continuous Android Studio Gradle compilation), and 20GB of free disk storage for SDK system images and project files.
- **Student/Teacher Smart Terminals:** Physical Android smartphones containing built-in cellular network modules (4G/5G) or Wi-Fi adapters alongside hardware-level GPS/GNSS sensors for precision multi-satellite coordinate capturing[1].
- **Physical Testing Peripherals:** At least two physical Android devices are mandatory to perform verification routines, as virtual emulators cannot accurately replicate continuous background hardware geofencing, hardware device binding hooks, or multi-satellite cellular drift[6].

2.3.2 Software Requirements:

- **Client Operating System:** Android 8.0 (API Level 26 - Oreo) or higher to support foreground service notification channels, high-accuracy privacy runtime location models, and standard background handler loops[3].
- **Cloud Services Configurations:** An active Google Firebase project provisioned on the Spark Free Tier, utilizing strict server-side Firestore Security Rules to prevent client-side document injection or data manipulation within the "attendance" collection[2].
- **Core APIs & Services:** Google Play Services framework actively running on client nodes, with precise (ACCESS_FINE_LOCATION) and background (ACCESS_BACKGROUND_LOCATION) tracking permissions explicitly granted by the user[1].

CHAPTER 3

DATABASE ANALYZING, DESIGN AND IMPLEMENTATION

For the backend persistence layer, a cloud-native, NoSQL document-oriented paradigm via Google Cloud Firestore was selected over traditional Relational Database Management Systems (RDBMS)[7]. Traditional relational databases enforce rigid schemas, tabular mappings, and high-overhead computational joins that introduce transactional latency during high-concurrency client updates, such as hundreds of students checking in simultaneously[7].

Cloud Firestore offers seamless horizontal scaling, low-latency document lookups, and native real-time websocket synchronization streams[7]. Data is organized into hierarchical Collections, which act as containers for standalone Documents[7]. Each document contains key-value pairs structured in JSON-like maps, supporting flexible data nesting perfectly tailored for asynchronous mobile applications[7].

3.1 Logical Database Schema Design (Data Models)

The logical structural mapping of the collection hierarchy implemented inside the Firebase Console is organized into the following document models:

1. users Collection

Stores centralized user profile attributes initialized during the primary registration phase[7].

- **Document ID:** Alpha-numeric User Unique Identifier (UID generated by Firebase Auth).
- **Fields:**
 - name: String (User's full name)
 - email: String (Institutional email address)
 - role: String ("Student" or "Teacher")

- deviceId: String (Unique hardware hash code for immutable 1:1 hardware terminal binding)[5].

2. attendanceCodes Collection

Manages temporary rolling verification session tokens generated dynamically by instructors.

- **Document ID:** Auto-generated unique string reference[7].
- **Fields:**
 - code: String (Randomized 4-digit numeric verification pin)
 - teacherId: String (UID link referencing the hosting instructor)
 - class: String (Target classroom section mapping, e.g., CSE-A)
 - subject: String (Active course title string)
 - startTime: String (Session initiation hour marker)
 - endTime: String (Session closure hour marker)
 - status: String (Operational session state: "active" or "expired")
 - expiryTime: Number (Timestamp defining the validity threshold of the pin).

3. attendance Collection

Maintains real-time valid presence logs for ongoing instructional lectures[7].

- **Document ID:** Auto-generated reference generated during the successful check-in transaction.
- **Fields:**
 - studentId: String (UID mapping of the authenticated student node)
 - teacherId: String (UID mapping of the validating instructor node)
 - subject: String (Active structural course code)
 - class: String (Target classroom section identifier)
 - startTime: String (Session start boundary timestamp)

- endTime: String (Session end boundary timestamp)
- date: String (Calendar validation index tracked as YYYY-MM-DD)

4. timetables / class_timetable Collection

Stores static baseline institutional weekly instructional configurations against which active session initiations are programmatically cross-referenced[7].

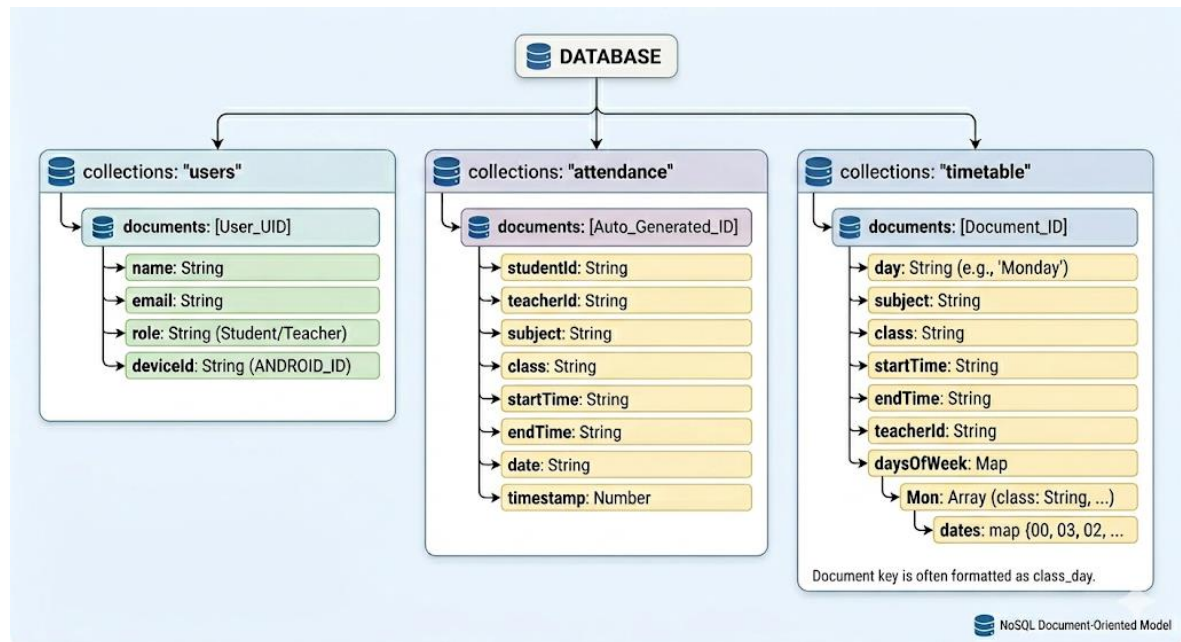


Fig. 3.1: Database schema design

3.2 Database Implementation and Security Policies

Database interactions are handled asynchronously using the native Google Firebase Android SDK. Write operations follow transactional integrity rules executed directly at the client application nodes.

To ensure complete data isolation and protect against structural payload injection, the cloud backend deploys server-side Firestore Security Rules[2]. These rules dynamically evaluate the user's role and device payload parameters before allowing operations on the collections. For instance, write requests to the attendanceCodes collection are rejected if the authenticated user's profile role does not match "Teacher"[2].

Student Check-In Flow (Simplified)

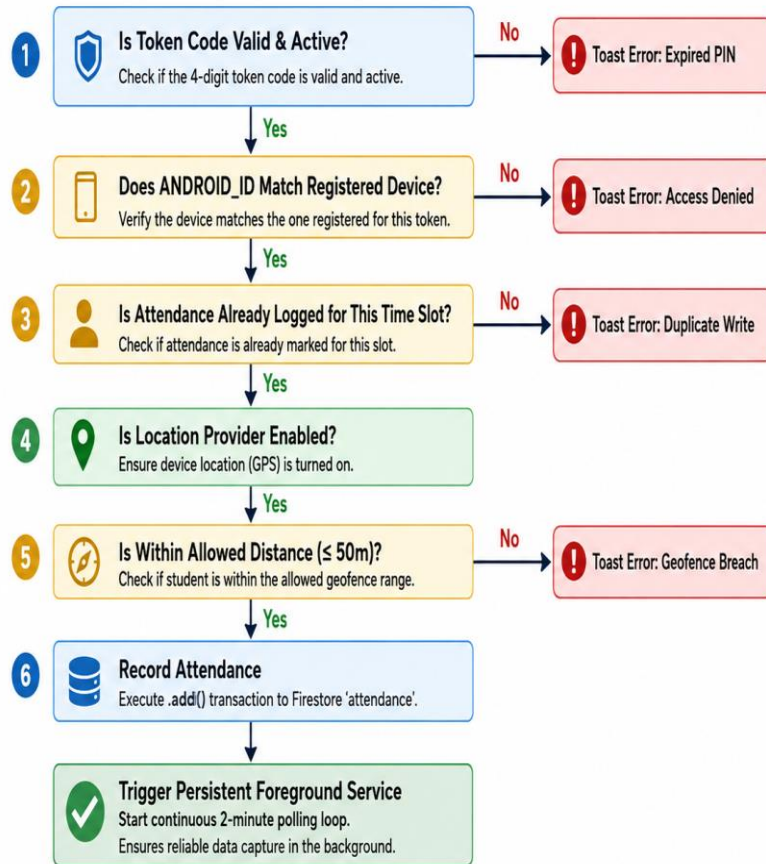


Fig. 3.2: Student Check-In-Flow

3.3 High-Level Design (HLD) Architectural Blueprint

The system architecture implements a decoupled, event-driven Client-Server architecture utilizing a cloud-native backend. The system is built with high horizontal scalability to manage concurrent transactions when hundreds of nodes submit location datasets simultaneously.

The global architecture comprises three major tiers:

- **Presentation Tier (Client Applications):** Native Android applications running isolated interfaces optimized for specific user identities (StudentActivity and TeacherActivity). These clients interact with device hardware layers to extract satellite telemetry via Google Play Services and system identification parameters via the OS kernel framework[1][5].

- **Application Gateway Layer:** Orchestrated by Firebase Services. It manages cryptographic session management via Firebase Authentication token emissions and handles incoming socket connections from client apps.
- **Persistence Tier (Data Store):** A dynamic, distributed NoSQL schema hosted via Google Cloud Firestore[7]. Data syncs in real-time through bidirectional WebSockets, pushing instant dashboard schema mutations across authenticated nodes without polling overhead[7].

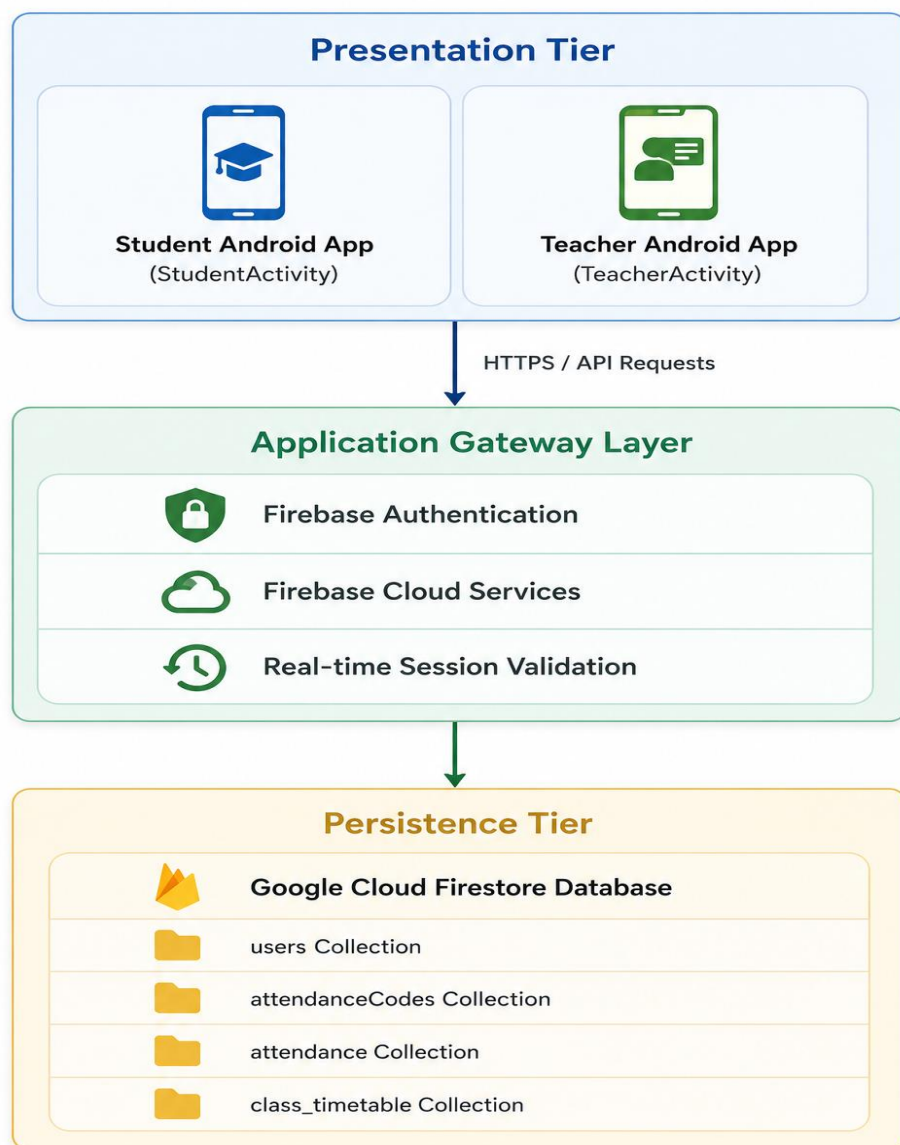


Fig. 3.3: High-Level Design

3.4 Low-Level Design (LLD) and Structural Abstractions

The structural framework of the system relies on an object-oriented paradigm designed around distinct functional subsystems. To ensure data validity and process separation, components are decoupled into distinct presentation layers, long-running processes, and localized data models.

3.4.1 Component Structural Relationships

The functional ecosystem maps operations to a strict object model:

- **Identity Management Domain:** MainActivity acts as the system coordinator. It verifies login tokens and interfaces with Settings.Secure to query hardware keys[5]. If valid, it instantiates user sessions.
- **Session Broadcasting Domain:** Orchestrated by TeacherActivity, which polls cell towers and satellites to establish an anchor coordinate point (Location instance). It binds temporal variables into a dictionary payload and commits a new record into the attendanceCodes collection[7].
- **Geospatial Verification Domain:** Built inside StudentActivity. It parses user inputs, evaluates session parameters from the cloud, and invokes Location.distanceBetween(double, double, double, double, float[]) to evaluate physical distance vectors[4].
- **Persistent Monitoring Domain:** Executed via AttendanceTrackingService (Foreground Engine) or PresenceWorker (Jetpack Background Handler). It sets up a persistent loop that maintains verification checks through background tasks independent of active UI threads[3].

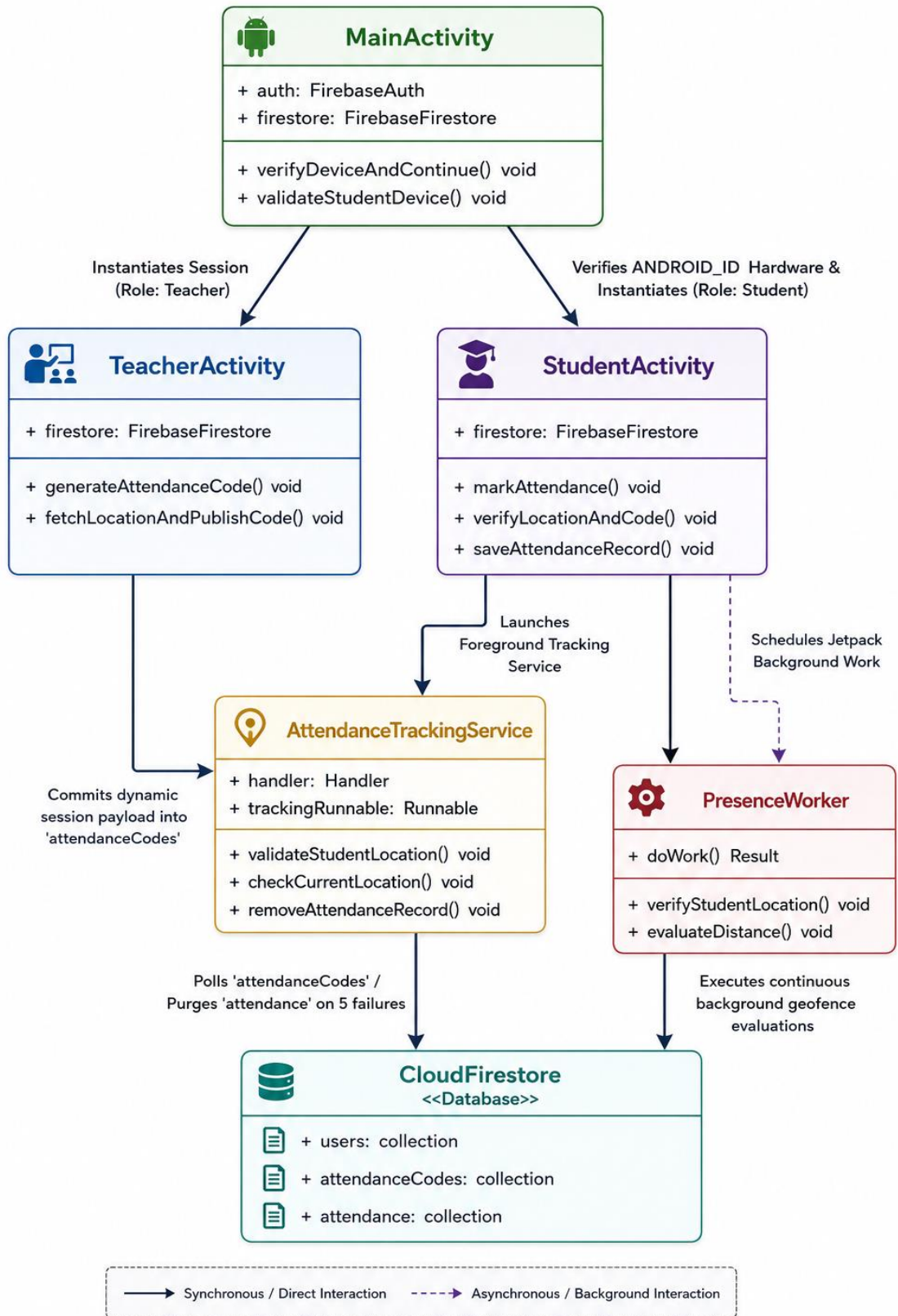


Fig. 3.4: Low-Level Design

CHAPTER 4

PROGRAM'S STRUCTURE ANALYZING AND GUI CONSTRUCTING

4.1 System Workflow and Architecture

The Smart Anti-Proxy Attendance System follows a cloud-based client-server architecture designed to ensure secure and reliable attendance management. The workflow begins when the user logs into the Android application using Firebase Authentication. During the authentication process, the system retrieves the device's unique `ANDROID_ID` and compares it with the device ID stored in the Cloud Firestore database[5]. This hardware binding mechanism ensures that one account can only be accessed from one registered device, thereby preventing proxy attendance through credential sharing[6].

After successful verification, the instructor creates an attendance session through the Teacher Module. The application generates a random 4-digit attendance code and stores the session details, including subject, classroom, and timing information, inside the Firestore database[7]. Simultaneously, the teacher's live GPS coordinates are captured using the Google Fused Location Provider API to establish the classroom's geofence boundary[1].

Students enter the generated attendance code through the Student Module. Before marking attendance, the application fetches the student's real-time GPS coordinates and calculates the distance between the student and teacher using the `Location.distanceBetween()` method[1][4]. Attendance is recorded only if the student is located within the predefined 50-meter radius[4].

To maintain continuous presence verification, the application activates a Foreground Service that performs location validation every two minutes in the background[3]. If the student exits the authorized geofence area for five consecutive checks, the system automatically removes the attendance entry from the Firestore database[2]. This workflow ensures real-time monitoring, eliminates fake attendance practices, and improves overall attendance authenticity[6].

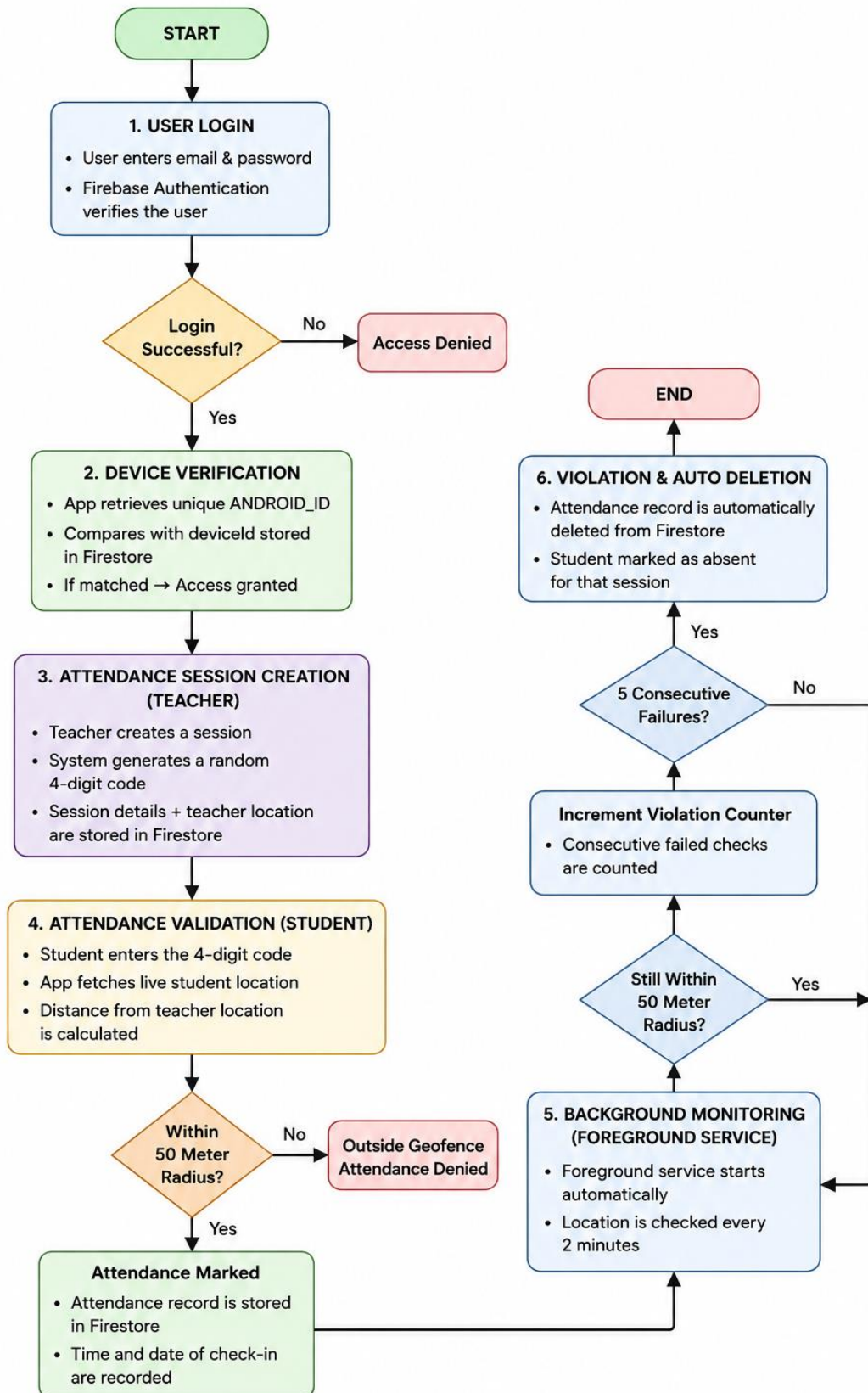


Fig. 4.1: System Workflow

4.2 User Authentication and Device Binding Module

The User Authentication and Device Binding Module provides secure login access and prevents proxy attendance. The system uses Firebase Authentication for user login and registration using email and password credentials.

After successful login, the application retrieves the device's unique `ANDROID_ID` using the Android Settings.Secure framework[5]. This device ID is stored in Cloud Firestore and linked with the user's account[7]. During future logins, the system compares the current device ID with the stored value[5]. If both match, access is granted; otherwise, login is denied[6].

This mechanism ensures that one account can only be used on one registered device, reducing unauthorized access and proxy attendance attempts[6]. The module also supports role-based access for both students and teachers.

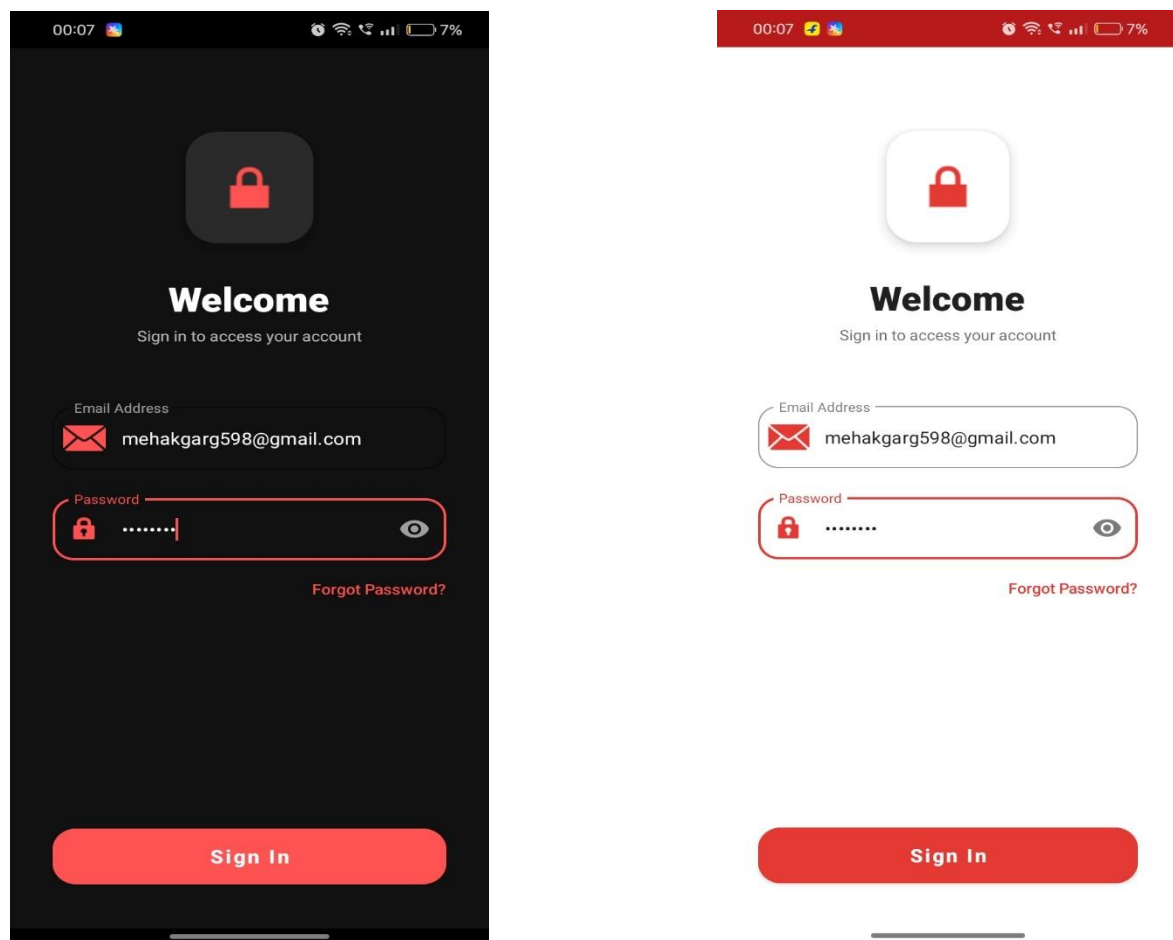


Fig. 4.2: User Authentication (Black or White Theme)

4.3 Teacher Dashboard Interface

The Teacher Dashboard allows instructors to manage attendance sessions efficiently. Teachers can create new attendance sessions by entering subject details, class information, and lecture timings. After the session is created, the system automatically generates a random 4-digit attendance verification code that is shared with students.

The dashboard also allows teachers to monitor attendance activities in real time and manage active classroom sessions securely[7].

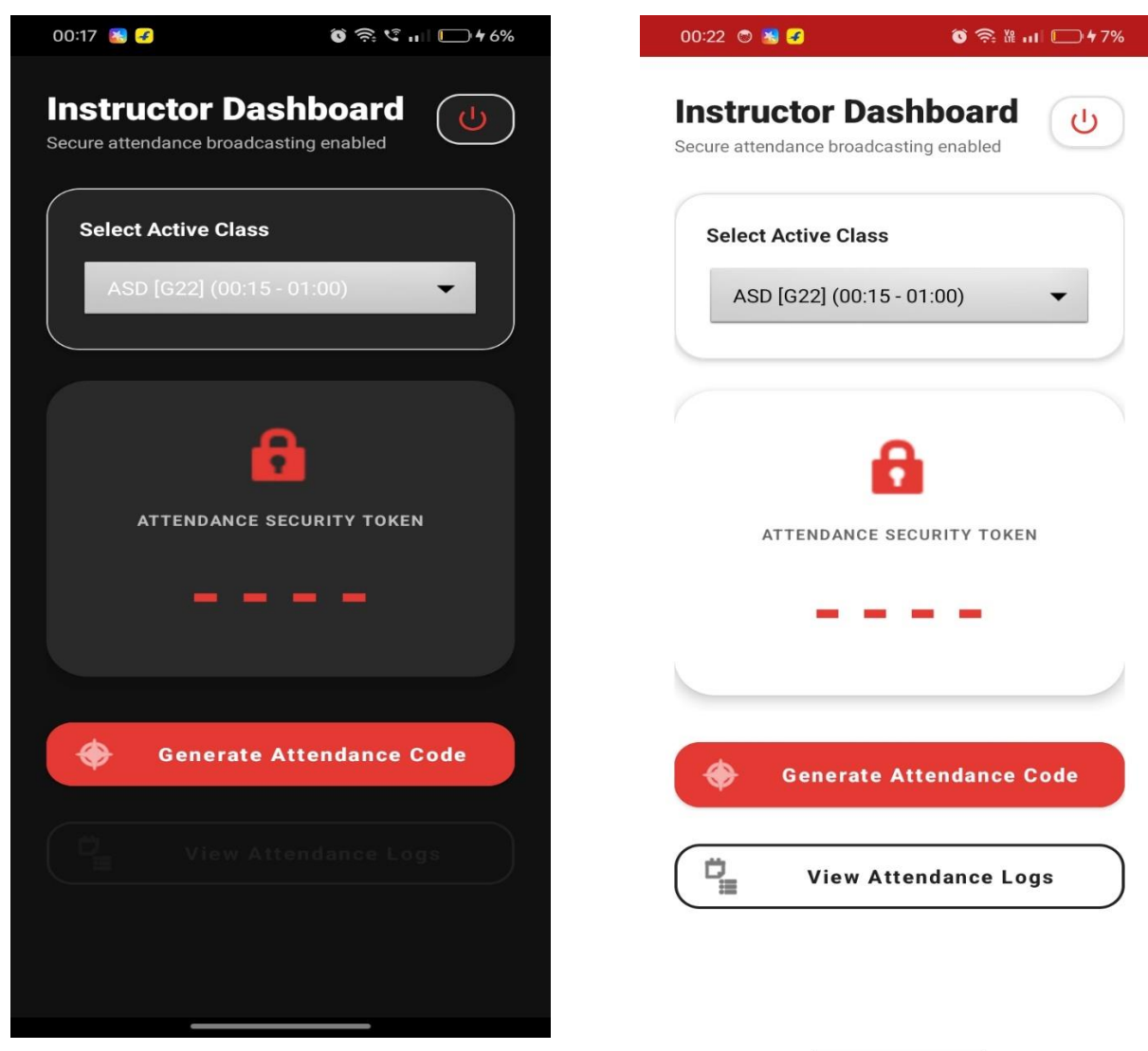


Fig. 4.3: Teacher Dashboard (Black or White Theme)

4.4 Student Dashboard Interface

The Student Dashboard enables students to submit attendance requests securely. Students enter the attendance verification code generated by the teacher through the application interface. After code submission, the system performs GPS-based location verification to confirm that the student is within the allowed classroom geofence radius[1].

If the location validation is successful, attendance is marked automatically[7]; otherwise, the request is rejected[2]. The interface is designed to provide a simple and user-friendly attendance experience.

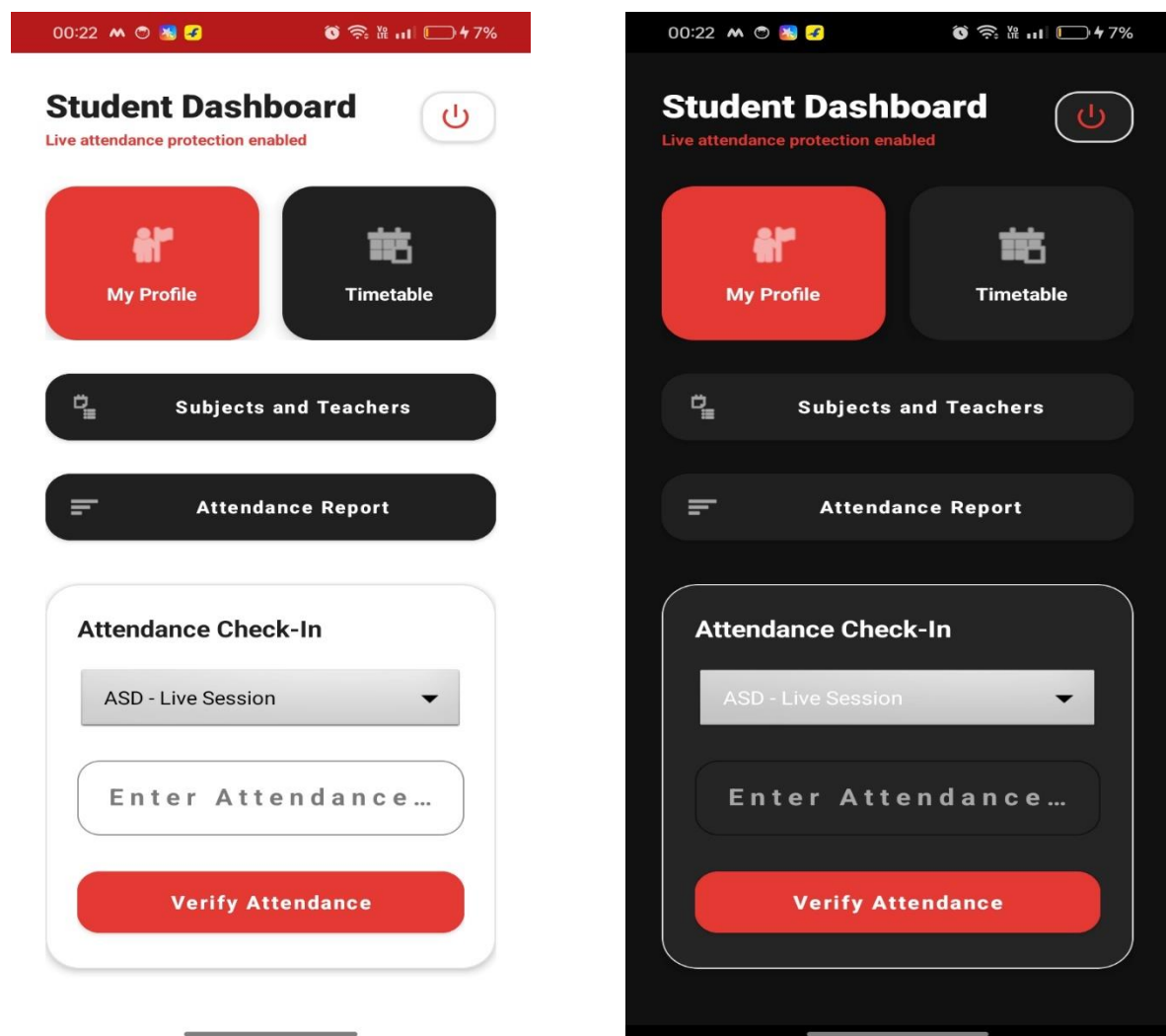
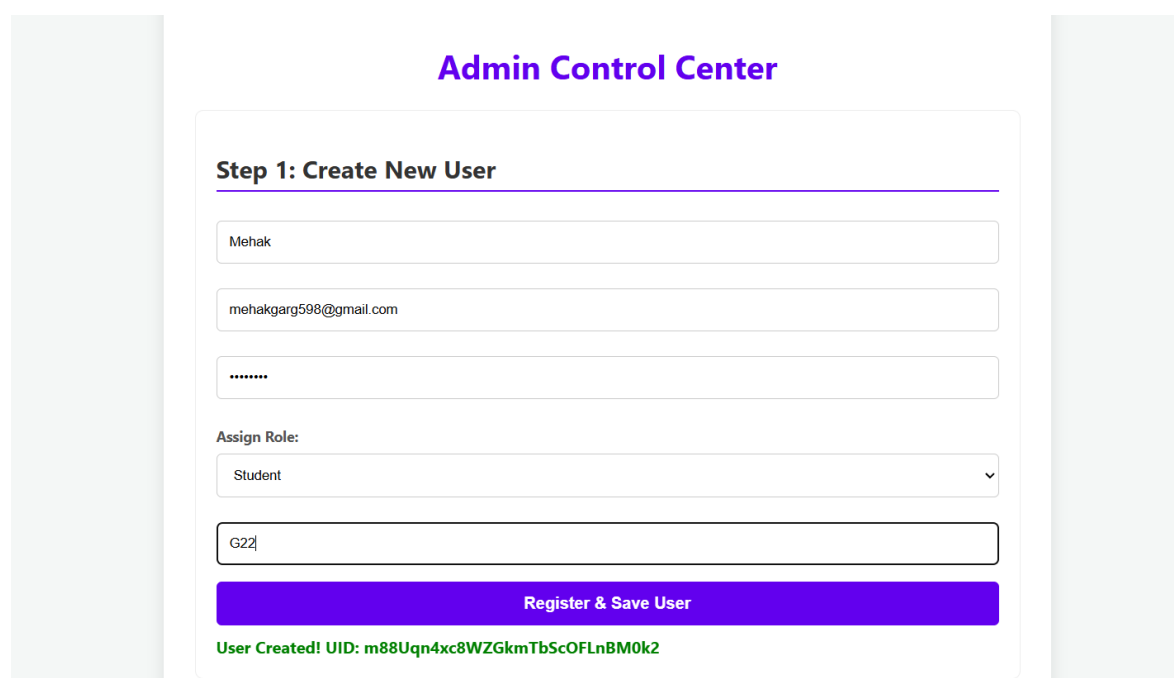


Fig. 4.4: Student Attendance (Black or White Theme)

4.5 Admin Dashboard Interface

The Admin Dashboard is designed to manage and monitor the overall attendance system. The interface provides centralized control for viewing registered users, attendance records, active sessions, and database activities. The admin panel developed using admin.html allows administrators to access system information through a web-based interface connected with Cloud Firestore[7].

The dashboard displays student and teacher details, attendance logs, and active classroom sessions in real time[7]. Administrators can also monitor database updates, verify attendance records, and manage user-related information efficiently. The interface provides a simple and organized structure for maintaining system transparency and administrative control.



The image shows a web interface titled "Admin Control Center" in purple. Below the title is a section "Step 1: Create New User" with a purple underline. The form contains several input fields: a text field with "Mehak", an email field with "mehakgarg598@gmail.com", a password field with "*****", a dropdown menu for "Assign Role:" with "Student" selected, and a text field with "G22". At the bottom is a purple button labeled "Register & Save User". Below the button, a green message states "User Created! UID: m88Uqn4xc8WZGkmTb5cOFLnBM0k2".

Fig. 4.5: Admin Dashboard (Create New User)

Step 2: Set Teacher Timetable

Assign classes with duration to a teacher.

Teacher's Email:

rajdavinder.singh@chitkara.edu.in

Select Day:

Thursday

Subject Name:

ASD

Start Time:

10 : 00

End Time:

11 : 00

Target Class:

G22

Update Teacher Timetable

Updated: ASD (10:00 - 11:00)

Fig. 4.6: Admin Dashboard (Set Teacher Timetable)

Step 3: Master Class Timetable

G22

Friday

FSE

FSE

CC-II

CC-II

01:00 - 02:00 PM: LUNCH BREAK

ADS

ADS

Save Master Schedule

Master Schedule Updated!

Fig. 4.7: Admin Dashboard (Student Timetable)

4.6 Student Profile Page

The Student Profile Page provides personalized information related to the authenticated student account. The interface displays important details such as student name, email address, class section, and registered device information retrieved from Cloud Firestore[7].

The profile page also allows students to view their attendance-related information and verify their registered account credentials. Device binding information linked through the ANDROID_ID is maintained securely to ensure that the account is accessed only from the authorized device[5][6].

The interface is designed with a simple and user-friendly layout to provide easy navigation and secure profile management for students.

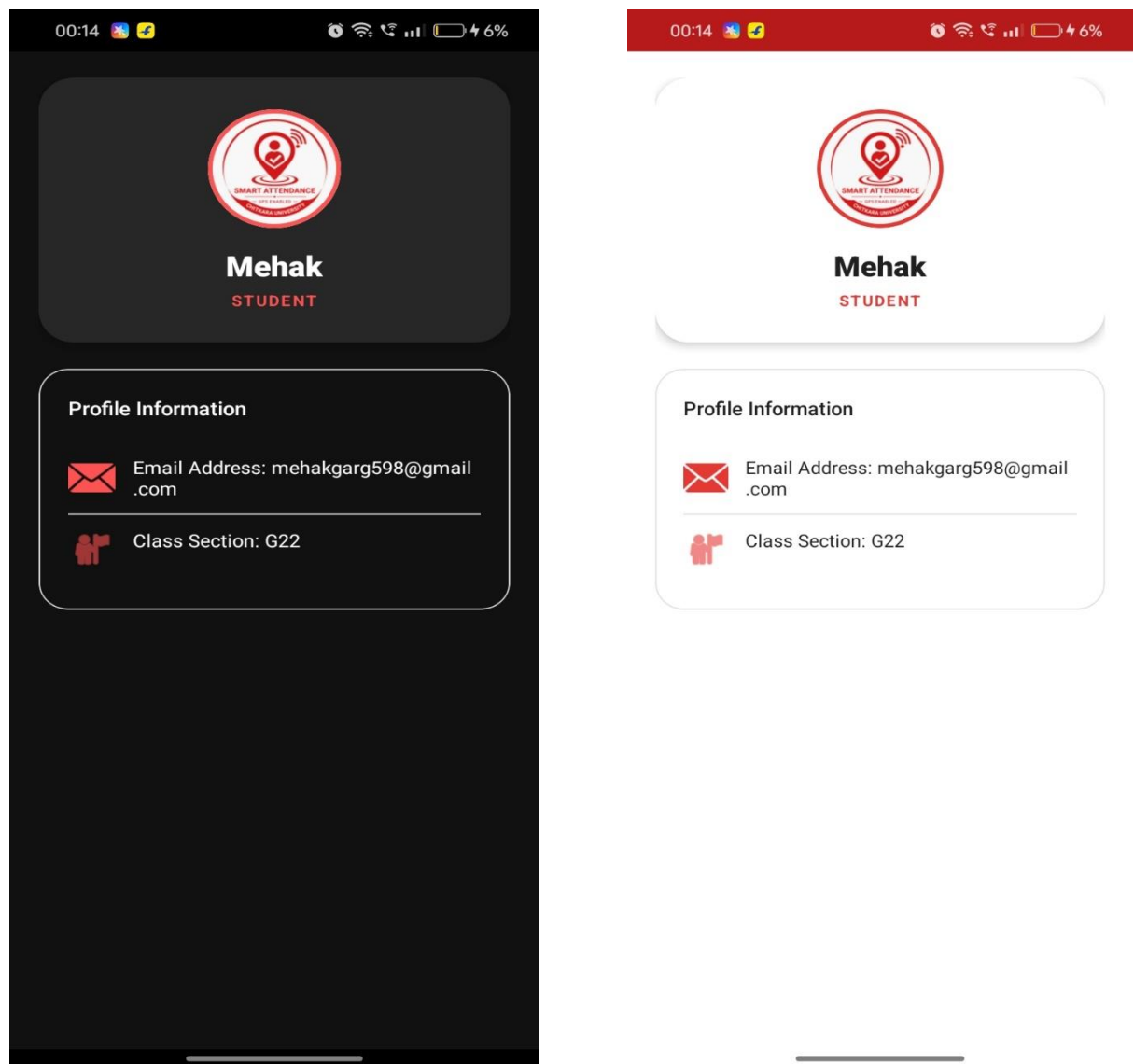


Fig. 4.6: Student Profile Management Interface

CHAPTER 5

CODE-IMPLEMENTATION AND DATABASE CONNECTIONS

This chapter focuses on the actual programmatic logic, core structural classes, and backend integration layers of the Smart Anti-Proxy Attendance System. To construct an operational infrastructure capable of mitigating proxy attendance vectors, the client-side Android application must execute continuous, asynchronous real-time communications with the Google Cloud Firestore persistence layer[7]. The sections below delineate the functional workflows, specifically unique device-hardware mapping, real-time geofenced proximity validation constraints, and the persistent multi-tiered background verification tracking service loop.

5.1. Device Hardware Binding and Authentication Logic

To neutralize identity manipulation where an individual logs into the system using a peer's authentication credentials from an off-site location, the platform enforces an absolute hardware-locking protocol[6]. Upon a successful authentication trigger via Firebase Auth, the application runtime intercepts the device's hardware abstraction layer to query a unique alphanumeric signature using the Settings.Secure.ANDROID_ID framework api.

The runtime engine performs an asynchronous lookup query against the target profile mapping inside the "users" collection in Cloud Firestore[7]. If a primary login instance is identified (null hardware field), the device signature string is permanently bound to that unique student record schema[7]. For all subsequent validation entries, the app executes an inline string comparison check[5]. If a mismatch occurs, the session context is forcefully terminated, successfully restricting one authenticated student account to a single authorized physical smartphone node[6].

The core programmatic logic executing this hardware-verification sequence inside MainActivity.java is implemented below:

```

usage
private void validateStudentDevice(String userId,
                                   String currentDeviceId,
                                   DocumentSnapshot userDocument) {
    firestore.collection(collectionPath: "users") CollectionReference
    .whereEqualTo( field: "deviceId", currentDeviceId) Query
    .get() Task<QuerySnapshot>
    .addOnSuccessListener( QuerySnapshot querySnapshot -> {
        if (!querySnapshot.isEmpty()) {
            for (DocumentSnapshot document :
                querySnapshot.getDocuments()) {
                if (!document.getId().equals(userId)) {
                    Toast.makeText(
                        context: MainActivity.this,
                        text: "This device is already linked to another account",
                        Toast.LENGTH_LONG
                    ).show();
                    return;
                }
            }
        }
        handleDeviceRegistration(
            userDocument,
            currentDeviceId
        );
    });
}

```

Fig. 5.1: MainActivity.java

5.2. Session Initialization and Proximity Validation

The real-time transactional monitoring pipeline is initialized by an instructor node utilizing the interface mechanics within TeacherActivity.java. When a tracking window is initiated, the application utilizes the high-precision FusedLocationProviderClient provider API to pull the instructor's baseline geographic coordinates[1]. Simultaneously, it maps the runtime temporal parameter against the master institutional timetable matrix inside the "timetable" collection[7]. Upon success, a randomized 4-digit token is generated and pushed to the cloud under the "attendanceCodes" collection with an active state string identifier and an epoch validation boundary (expiryTime)[7].

To mark their presence, student terminals invoke StudentActivity.java, where they feed the session code into the client interface. The submission process executes three mandatory verification filters:

1. **Active Session Validation:** The client queries the "attendanceCodes" collection to evaluate token parity, verify an active conditional status, and ensure the local system clock has not crossed the session's upper epoch expiration boundary[7].

2. **Duplicate Transaction Filtering:** To prevent repetitive transactional commits within a single session block, the client queries the "attendance" collection utilizing an composite filter array composed of studentId, the calendar date index YYYY-MM-DD, and active lecture time bounding boxes[7]. If a document match occurs, the submission loop is broken.
3. **Geospatial Distance Computing:** Once the token validation and duplicate filters are cleared, the application reads the student's live location vectors and invokes the native framework utility `Location.distanceBetween()` to calculate linear displacement relative to the instructor's baseline anchor coordinates[2][7].

If the computed linear displacement satisfies the perimeter rule ($\$Distance \leq 50 \text{ meters}$), a transactional document write containing structural state logs (studentId, teacherId, subject, class, startTime, endTime, date, timestamp) is committed to the database.

The data storage logic for a successful check-in transaction inside `StudentActivity.java` is implemented below:

```

1 usage
private void saveAttendanceRecord(Map<String, String> sessionData) {
    if (auth.getCurrentUser() == null) return;

    Map<String, Object> attendanceRecord = new HashMap<>();
    attendanceRecord.put("studentId", auth.getCurrentUser().getUid());
    attendanceRecord.put("teacherId", sessionData.get("teacherId"));
    attendanceRecord.put("subject", sessionData.get("subject"));
    attendanceRecord.put("class", sessionData.get("class"));
    attendanceRecord.put("startTime", sessionData.get("startTime"));
    attendanceRecord.put("endTime", sessionData.get("endTime"));
    attendanceRecord.put("date", new SimpleDateFormat(pattern: "yyyy-MM-dd", Locale.getDefault()).format(new Date()));
    attendanceRecord.put("timestamp", System.currentTimeMillis());

    firestore.collection(collectionPath: "attendance") CollectionReference
    .add(attendanceRecord) Task<DocumentReference>
    .addOnSuccessListener(DocumentReference documentReference -> {
        Toast.makeText(context: this, text: "Attendance submitted successfully", Toast.LENGTH_LONG).show();
        codeInput.setText("");

        startAttendanceTracking(sessionData, documentReference.getId());
        loadTodayClasses();
    })
    .addOnFailureListener(Exception e -> Toast.makeText(context: this, text: "Failed to save attendance", Toast.LENGTH_SHORT).show());
}

```

Fig. 5.2: StudentActivity.java

To ensure session integrity and eliminate location spoofing or post-validation classroom departure, the platform features a continuous background verification architecture. Instead of relying solely on a singular entry-point location check, the system implements a persistent monitoring cycle that validates student proximity throughout the designated lecture duration.

The tracking ecosystem can be engineered around two distinct Android architecture components, each tailored to specific operational lifecycles:

- **Foreground Service (AttendanceTrackingService):** Functions as the primary runtime engine during an active application state[3]. Implemented as a persistent system component utilizing a mandatory notification channel, it prevents the Android operating system from reclaiming allocated memory assets when the user minimizes or backgrounds the application.
- **Persistent Background Worker (PresenceWorker):** Leverages the Android Jetpack WorkManager API to handle background tasks independent of the main application thread lifecycle[3]. Utilizing an asynchronous, self-scheduling execution routine, it handles deferred background database synchronization tasks without introducing UI layer latency or execution bottlenecks[3].

The code implementing this anti-escape background tracking loop inside PresenceWorker.java is structured below:

```

1 usage
private void evaluateDistance(Location location, double targetLatitude, double targetLongitude, String attendanceDocumentId) {
    float[] results = new float[1];
    Location.distanceBetween(
        location.getLatitude(),
        location.getLongitude(),
        targetLatitude,
        targetLongitude,
        results
    );
    float distance = results[0];
    firestore.collection( collectionPath: "users") CollectionReference
        .document(studentId) DocumentReference
        .get() Task<DocumentSnapshot>
        .addOnSuccessListener( DocumentSnapshot userDocument -> {
            long failureCount = userDocument.contains("consecutiveFailures") ? userDocument.getLong( field: "consecutiveFailures") : 0;
            if (distance > MAX_ALLOWED_DISTANCE) {
                handleAttendanceFailure(
                    failureCount,
                    attendanceDocumentId
                );
            } else {
                resetFailureCounter();
                Log.d(TAG,
                    msg: "Student verified within allowed range");
            }
            scheduleNextVerification();
        });
}

```

Fig. 5.3: PresenceWorker.java

5.4 Database Synchronization Process

The Smart Anti-Proxy Attendance System utilizes Cloud Firestore for real-time database synchronization and cloud data management[7]. All attendance-related operations, including session creation, attendance marking, device verification, and attendance deletion, are synchronized automatically between the Android client application and the Firestore database[7].

Whenever a teacher creates a new attendance session or a student marks attendance, the updated data is instantly reflected in the cloud database without requiring manual refresh operations[7]. Firestore's real-time synchronization mechanism enables continuous data consistency across all authenticated devices connected to the system[7].

The application also performs automatic cloud synchronization for background monitoring activities. If a student violates the geofence boundary continuously, the Foreground Service automatically updates or removes the attendance record from Firestore in real time[2]. This ensures that attendance records remain accurate, secure, and continuously updated throughout the active session.

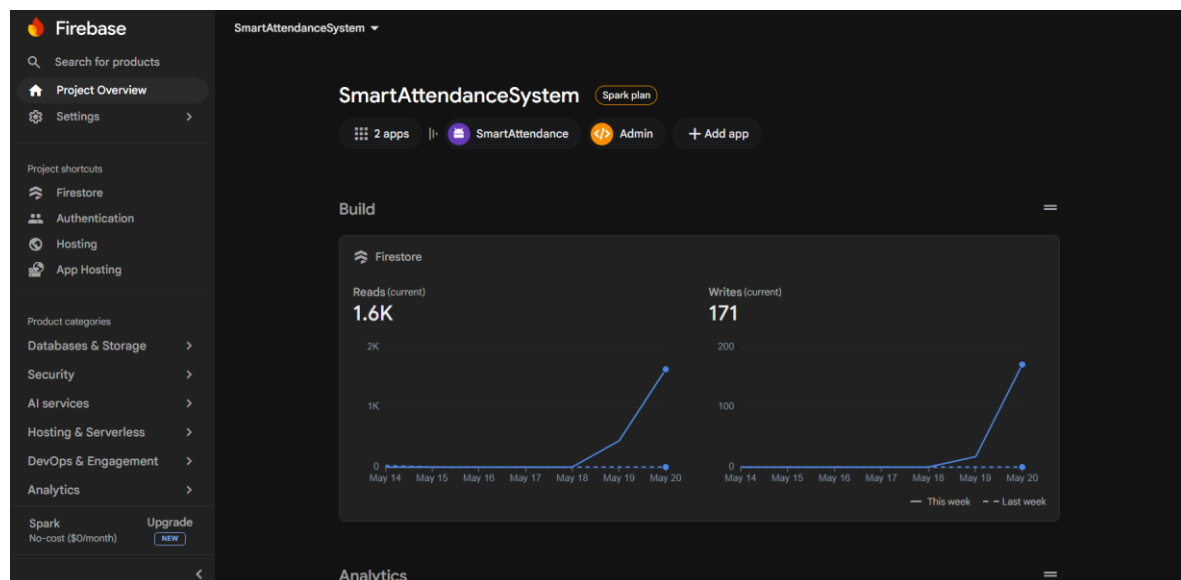


Fig. 5.4: Cloud Firestore Real-Time Database Synchronization

5.5 Logout-Based Attendance Enforcement

To further strengthen attendance authenticity, the system introduces a logout-triggered absence marking mechanism[6]. This ensures that students cannot simply mark attendance and exit the session early without consequences.

When a student explicitly logs out during an active live class session, the application first checks the current session state from the "attendanceCodes" collection and verifies whether the lecture is still ongoing[7]. If the session is active, the system treats the logout event as a violation of continuous presence policy[6].

Upon logout trigger, the following backend operations are executed:

The system fetches the latest attendance record associated with the `studentId`, `teacherId`, and active session timestamp[7].

- A conditional update is performed in Firestore where the attendance status is immediately marked as "ABSENT" or flagged as "INVALID_SESSION" depending on institutional policy configuration[2][7].
- If continuous tracking was active via `AttendanceTrackingService` or `PresenceWorker`, the service is forcefully terminated and a final state check is executed before shutdown[3].
- An optional audit log entry is created to record that the student exited the system during an active lecture window[7].

This mechanism ensures that attendance integrity is maintained even at the session exit level and prevents users from bypassing geofence monitoring by simply logging out after initial check-in[6].

```

private void logoutStudent() {
    if (auth.getCurrentUser() == null) { return; }
    String userId = auth.getCurrentUser().getUid();
    String todayDate = new SimpleDateFormat( pattern: "yyyy-MM-dd", Locale.getDefault()).format(new Date());
    Toast.makeText( context: this, text: "Ending current session", Toast.LENGTH_SHORT).show();
    String selectedClass = teacherSpinner.getSelectedItem() != null ? teacherSpinner.getSelectedItem().toString() : "";
    Map<String, String> sessionData = sessionDataMap.get(selectedClass);
    if (sessionData != null) { String currentSubject = sessionData.get("subject");
        String classSection = sessionData.get("class");
        String startTime = sessionData.get("startTime");
        String endTime = sessionData.get("endTime");
        firestore.collection( collectionPath: "attendance").whereEqualTo( field: "studentId", userId)
            .whereEqualTo( field: "date", todayDate).whereEqualTo( field: "class", classSection)
            .whereEqualTo( field: "startTime", startTime)
            .whereEqualTo( field: "endTime", endTime).get().addOnSuccessListener( QuerySnapshot querySnapshot -> {
                for (DocumentSnapshot document : querySnapshot.getDocuments()) {firestore.collection( collectionPath: "attendance")
                    .document(document.getId()).delete(); }
            });
        stopService(new Intent( packageContext: this, AttendanceTrackingService.class));
        auth.signOut();
        Intent intent = new Intent( packageContext: StudentActivity.this, MainActivity.class);
        intent.setFlags(Intent.FLAG_ACTIVITY_NEW_TASK | Intent.FLAG_ACTIVITY_CLEAR_TASK);
        startActivity(intent);
        finish();
        Toast.makeText( context: this, text: "Logged out successfully. Active attendance removed.", Toast.LENGTH_LONG).show();
    }
}

```

Fig. 5.5:Logout-Based Enforcement

CHAPTER 6

SYSTEM TESTING

System testing was performed to verify the functionality, reliability, and security of the Smart Anti-Proxy Attendance System. Different testing scenarios were executed to validate key system features such as authentication, device verification, attendance generation, geofencing, and background monitoring.

6.1 Functional Testing

1. Functional testing was conducted to ensure all system modules worked according to requirements.
2. The login module was tested using valid and invalid credentials.
 - Only registered users were allowed access.
 - Unauthorized login attempts were successfully blocked using Firebase Authentication.
3. The device binding system was tested using registered and unregistered devices.
 - Device verification was done using ANDROID_ID[5].
 - Access from unauthorized devices was correctly denied[6].
4. The attendance session generation was tested via the Teacher Dashboard.
 - Random 4-digit attendance codes were generated successfully.
 - Session details were stored in Cloud Firestore in real time[7].
5. The student attendance verification module was tested with valid and invalid codes.
 - Correct codes allowed attendance marking.
 - Incorrect codes were rejected[7].
6. The geofencing feature was tested at different distances from the teacher's location.
 - Students within the 50-meter radius were marked present[4].
 - Students outside the radius were denied attendance[2].

7. The continuous monitoring system was tested using Android Foreground Service.
 - Students staying outside the geofence for more than 10 minutes had their attendance automatically removed[7].
 - Real-time updates between app and database worked successfully.
8. Overall testing confirmed that the system:
 - Prevents proxy attendance[6]
 - Ensures secure authentication
 - Maintains accurate attendance tracking[1]
 - Supports real-time monitoring during class sessions[7]

CHAPTER 7

LIMITATIONS

While the Smart Anti-Proxy Attendance System provides a secure, multi-layered validation mechanism, it operates under certain real-world technical and environmental limitations:

7.1. Indoor GPS Attenuation

- **Signal Loss:** GPS microwave signals experience significant attenuation when passing through thick concrete roofs and multi-story structures[1].
- **Coordinate Drift:** In deep indoor labs or basements, accuracy can drop beyond the 50-meter radius, occasionally causing false fence violation logs for students who are legitimately present[4].

7.2. OS-Level Background Restrictions

- **Device Fragmentation:** Different Android vendors (like Xiaomi, Samsung, OnePlus) implement aggressive background optimization algorithms.
- **Service Termination:** These power-saving policies can forcefully kill or suspend the underlying WorkManagerbackground loop, disrupting the continuous 2-minute presence checks[3].

7.3. High Battery and Network Dependency

- **Resource Consumption:** Triggering hardware GPS location sensors continuously every 2 minutes accelerates smart terminal battery drain[1].
- **Connectivity Needs:** The system requires stable 4G/5G mobile networks or campus Wi-Fi for immediate cloud sync; network timeouts can fail database transactions[7].

7.4. Vulnerability to Spoofing at Kernel Layer

- **Location Mocking:** Advanced users with rooted Android operating systems can bypass the system by injecting falsified coordinates via specialized location-spoofing developer tools.
- **Validation Failure:** If coordinates are manipulated at the OS-kernel level, the application's location receiver may capture the fake data as authentic, temporarily bypassing the geo-fencing constraint[4].

CHAPTER 8

CONCLUSION

The **Smart Anti-Proxy Attendance System** successfully delivers a modern, automated, and highly dependable alternative to vulnerable and resource-heavy manual attendance tracking methodologies. By implementing a multi-layered security architecture directly within a native Android ecosystem, the project fulfills all its core operational and engineering objectives. The validation matrix deployed—combining Firebase identity authentication, hardware-level device binding (ANDROID_ID), dynamic geo-fencing via the Google Fused Location Provider API, and automated background tracking via Android WorkManager effectively neutralizes the systemic proxy issues that affect traditional classroom management systems[1][3][6].

From a technical perspective, the application proves that software-driven location tracking and device verification can achieve the same levels of data integrity as expensive hardware installations like biometric or RFID terminals, but at a fraction of the cost and deployment time. Enforcing the strict **10-minute absence validation rule** ensures that student accountability is maintained continuously throughout the instructional window, rather than just during a single moment at the start of class.

While environmental bottlenecks like indoor GPS drifts and system-level background task limitations pose minor operational constraints, the current stable prototype functions with high precision in standard academic environments. In conclusion, this integrated project successfully provides an effective, highly scalable, and completely transparent tool that optimizes teaching hours, eliminates administrative inefficiencies, and protects structural data integrity for modern educational institutions[2][3][4][7].

CHAPTER 9

FUTURE SCOPE

The current prototype provides a secure foundation for automated attendance tracking. To scale this platform across a multi-department university campus, the following modules can be integrated in future updates:

9.1. Automated Proactive Notifications

By integrating Firebase Cloud Messaging (FCM), the application can monitor the background counter and send real-time alerts if a student exits the classroom radius[3]. Instant warnings like "Warning: You are outside the classroom zone. Return within 2 minutes to prevent an automatic absence" will actively protect students against localized indoor GPS drifts[1].

9.2. University ERP Portal Integration

Establishing a direct bridge with the university's central Enterprise Resource Planning (ERP) database is essential. By exposing secure, real-time APIs or cloud webhooks, the system can automatically push attendance logs directly into the official university portal, eliminating manual Excel management and streamlining eligibility evaluations[2][7].

9.3. BLE Beacon and Biometric Hybridization

To completely neutralize OS-level GPS spoofing and account sharing, the verification framework can be upgraded into a hybrid hardware-biometrics model[6]:

- **BLE Beacons:** Low-cost Bluetooth Low Energy beacons inside each lecture hall will introduce a dual-clearance rule, requiring the app to verify GPS data and detect the room's specific Bluetooth signal[4].
- **On-Device Biometrics:** Native fingerprint verification right before code entry ensures that the bound physical device is operated exclusively by its authorized owner[6].

CHAPTER 10

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